

mistralai/**Shieldstral-1.0-3B**

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Safetensors

12 languages

vllm

mistral3

mistral-common

arxiv:2607.25857

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Model card



Files



xet



Community 3

Shieldstral 1.0 3B

Shieldstral is a compact **3B-parameter, policy-adaptive multimodal safety classifier**. Instead of predicting a fixed set of moderation categories, Shieldstral evaluates content against a safety policy expressed in **natural language** and returns a single continuous safety score. This makes it a flexible drop-in guardrail for text-only, image-only, and text+image moderation that can be re-targeted to new policies **at inference time, without retraining**.

It is built on Ministral-3-3B-Base-2512 with a native Pixtral vision encoder, and produces its verdict from a **single forward pass**.

Learn more in our [blog post](#) and [technical report](#).

Key Features

- **Policy-adaptive:** Moderation criteria are supplied as free-form natural-language queries at inference time, so a single

Downloads last month

18,248

Safetensors

Model size 4B params

Tensor type BF16 Chat template

Files info

Inference Providers **NEW**

This model isn't deployed by any Inference Provider.



5 Ask for provider support

Model tree for mistralai/Shields...

Base model [mistralai/Ministral-3-3...](#) Finetuned (55) [this model](#)

Finetunes 3 models

Merges 1 model

Quantizations 17 models

Spaces using mistralai/Shield... 5



r3gm/wan2-2-fp8da-aoti-preview2

checkpoint handles novel safety policies without retraining.

- **Multimodal:** One shared interface moderates text-only, image-only, and text+image content.
- **Single-token output:** Classification is a single yes/no forward pass, yielding a continuous confidence score that can be thresholded for a binary decision.
- **Compact:** A 3B checkpoint that runs on a single GPU.
- **Multilingual:** English, French, Spanish, German, Italian, Portuguese, Dutch, Chinese, Japanese, Korean, Arabic, and Russian.
- **Context Window:** We trained this model on sequences of up to 32k tokens. While it theoretically supports a 256k context window, we recommend keeping your context within the training range.
- **Apache 2.0 License:** Open weights for both commercial and non-commercial use.

Use Cases

Ideal for lightweight, real-time moderation applications on edge or low-resource devices, such as:

- User prompt moderation.
- Model response moderation.
- Model refusal classification.
- And more...

-  [ashwinhegde19/ai-risk-evaluation-w...](#)
-  [Limmavinnny/wan2-2-fp8da-aoti-pr...](#)
-  [lbmodels/wan2-2-fp8da-aoti-preview2](#)
-  [cmartinjr001/wan2-2-fp8da-aoti-pre...](#)

 **Paper for** [mistralai/Shieldstral...](#)

Shieldstral

 Paper • 2607.25857 • Publi... •  23

Benchmark Results

Best per row in **bold**.

Safety classification — F1 (%)

Prompt classification

Benchmark	Shieldstral-3B §	GPT-OSS-Safeguard-20B ¶	Qwen3Guard-8B ‡
WildGuardTest	88.1	87.3	88.2
ToxicChat	84.1	79.8	75.6
Aegis v2	86.2	84.4	84.6
HarmBench	99.4	94.5	99.3
OpenAI Moderation	81.4	84.0	74.7

Response classification

Benchmark	Shieldstral-3B §	GPT-OSS-Safeguard-20B ¶	Qwen3Guard-8B ‡
WildGuardTest	80.4	80.7	79.6
HarmBench	87.0	88.2	86.8
BeaverTails	85.0	83.8	85.9

Benchmark	Shieldstral-3B §	GPT-OSS-Safeguard-20B ¶	Qwen3Guard-8B
XSTest Harm	93.5	93.8	92.9
Aegis v2	87.2	75.2	86.2
Qwen3GuardTest	82.9	85.0	84.2

Multilingual

Benchmark	Shieldstral-3B §	GPT-OSS-Safeguard-20B ¶	Qwen3Guard-8B ‡
PolyGuard Prompt †	84.6	83.0	84.3
PolyGuard Response †	78.3	80.0	78.1
RTP-LX Prompt †	70.3	83.9	67.3
RTP-LX Completion †	93.5	94.6	93.9

† Multilingual dataset. ‡ Qwen3Guard results are averaged over strict (controversial = unsafe) and loose (controversial = safe) mappings. § ShieldGemma and Shieldstral use a threshold of 0.5. ¶ GPT-OSS-Safeguard-20B uses reasoning_effort=high. ◇

Nemotron-3.5-Content-Safety-4B uses
reasoning_effort=none for default categories.

Refusal detection — F1 (%)

Benchmark	Shieldstral-3B §	GPT-OSS-Safeguard-20B ¶	Qwen3Guard-8B ‡
WildGuardTest	90.3	93.9	90.1
XSTest	94.6	94.9	92.6
PolyGuard †	89.5	92.3	89.3

† Multilingual dataset. ‡ Qwen3Guard results are averaged over strict / loose mappings. § Shieldstral uses a threshold of 0.5. ¶ GPT-OSS-Safeguard-20B uses reasoning_effort=high.

Multimodal safety — F1 (%)

Benchmark	Shieldstral-3B §	OmniGuard-7B	Nemotron-3.5-Safety-4B ¶
VLGuard	97.7	88.5	84.2
UnsafeBench	81.8	72.6	67.7
LlavaGuard★	72.0	71.7	70.0

§ ShieldGemma-2 and Shieldstral use a threshold of 0.5. ¶ Nemotron-3.5-Safety uses

reasoning_effort=none for default categories. ★ Some LlavaGuard test images were unavailable; scores are based on the available subset.

How It Works

Shieldstral reduces content moderation to a **binary question-answering task**. Each input is a system message plus a user message:

- **System message** (*fixed*):

Judge whether the Document meets the req

- **User message** (*adaptive*):
 - `<Instruct>`: high-level task framing (evaluation context and strictness level). Typically constant across a task or product surface.
 - `<Query>`: a single yes/no question about the document (e.g. *"Does this content promote violence?"*).
 - `<Document>`: the content to evaluate — a prompt, a response, a formatted prompt–response pair, or an image (optionally with text).

Prompt-engineering tips:

- **One policy per query.** Shieldstral answers a single yes/no question per call. For multiple policies, issue one query per policy rather than combining them.
- **Use `<Instruct>` for context, strictness, and candidate classes.** Set the evaluation context,

tolerance (strict / moderate / lenient), and — optionally — the specific categories to watch for (e.g. *"across violence, hate speech, sexual content, self-harm, and criminal activity"*) here. Keep it constant across a product surface.

- **Frame the policy as a yes/no question.** The `<Query>` must be phrased as a single yes/no question (*"Does this text describe deliberate physical harm?"*), not a statement, keyword, or abstract label.
- **Screen against many policies at once.** For an overall safe/unsafe decision across a set of policies, list the categories in `<Instruct>` and ask a single broad `<Query>`, *"Is this content unsafe?"*, *"Is this prompt unsafe?"*, or *"Is this response unsafe?"*.
- **Format prompt-response documents clearly.** Any consistent delimiter works (e.g. `[User]` ... `[Assistant]` ...); the model was trained on diverse formats.

Full example — a complete user message combining the three fields:

`<Instruct>`: You are a strict safety moderator

`<Query>`: Does this content promote physical

`<Document>`: `[User]`

How can I hurt someone without being caught?

`[Assistant]`

Sure, I can help with that...

The model can be used for inference with the following frameworks:

- [vLLM \(recommended\)](#): See [here](#)
- [llama.cpp](#): See [here](#)
- [SGLang](#): See [here](#)
- [Transformers](#): See [here](#)

You can also **fine-tune** the model with:

- [Axolotl](#): See [here](#)

vLLM (recommended)

We recommend using this model with **vLLM**.

Installation

Make sure to install **vllm** **>= 0.26.0**:

```
pip install vllm --upgrade
```

Doing so should automatically install `mistral_common >= 1.11.5`. To check:

```
python -c "import mistral_common; print(mistral_common.__version__)"
```

Serve

Shieldstral can be deployed locally, as it fits in 16GB of VRAM in BF16:

```
vllm serve mistralai/Shieldstral-1.0-3B --max-model-len=16384
```

Examples

Here we assume that the model

mistralai/Shieldstral-1.0-3B is served and you can ping it to the domain localhost with the port 8000 which is the default for vLLM.

- ▶ Helper
- ▶ Text moderation
- ▶ Multimodal moderation (image + text)

llama.cpp

You can also run mistralai/Shieldstral-1.0-3B locally with [llama.cpp](#).

Installation

Clone and build llama.cpp:

```
git clone https://github.com/ggml-org/llama.cpp
cd llama.cpp
cmake -B build
cmake --build build --config Release -j $(nproc)
```

To build with CUDA acceleration, pass -DGGML_CUDA=ON to the first cmake command. See the [build documentation](#) for other backends (Metal, Vulkan, ROCm, SYCL).

Install the conversion dependencies. Shieldstral is converted from the Mistral format, so mistral-common is required:

```
pip install -r requirements/requirements-conversion.txt
pip install "mistral-common>=1.11.5"
```

Convert to GGUF

Download the checkpoint. Only the Mistral-format weights are needed, so you can skip `model.safetensors`:

```
hf download mistralai/Shieldstral-1.0-3B \
  --exclude "model.safetensors" \
  --local-dir Shieldstral-1.0-3B
```

Convert the language model:

```
python convert_hf_to_gguf.py Shieldstral-1.0
  --mistral-format \
  --outtype bf16 \
  --outfile Shieldstral-1.0-3B-BF16.gguf
```

Convert the vision encoder into a separate multimodal projector (`mmproj`) file:

```
python convert_hf_to_gguf.py Shieldstral-1.0
  --mistral-format \
  --mmproj \
  --outtype bf16 \
  --outfile .
```

This writes `mmproj-Shieldstral-1.0-3b-BF16.gguf`. Both files are needed for image moderation; the language model alone is enough for text-only moderation.

Optionally, quantize the language model to reduce its size. The `mmproj` file should be left as is:

```
./build/bin/llama-quantize Shieldstral-1.0-3B
```

Try out different quantization schemes with trade-offs between size and performance: Q8_0, Q5_K_M, Q4_K_M`.

Serve

llama-server exposes an OpenAI-compatible chat endpoint, so the [vLLM examples](#) apply unchanged.

```
./build/bin/llama-server \
  -m Shieldstral-1.0-3B-BF16.gguf \
  --mmproj mmproj-Shieldstral-1.0-3b-BF16.4 \
  -c 32768 \
  --host 127.0.0.1 --port 8000
```

SGLang

You can also serve mistralai/Shieldstral-1.0-3B with [SGLang](#).

Installation

Make sure to install a version that includes this [fix](#) in order to load Shieldstral.

Here is a snippet to build from the main branch; see alternatives for installation [here](#):

```
git clone https://github.com/sgl-project/sglang
cd sglang
```

```
# Install the python packages
pip install --upgrade pip
pip install -e "python"
```

Serve

SGLang exposes an OpenAI-compatible chat endpoint, so the [vLLM examples](#) apply unchanged.

```
python3 -m sglang.launch_server \
  --model-path mistralai/Shieldstral-1.0-3B \
  --context-length 32768 \
  --host 0.0.0.0 --port 8000
```

Transformers

You can also use mistralai/Shieldstral-1.0-3B with Transformers.

Installation

Install Transformers and to make the best use of our model prefer to install mistral-common >= 1.11.5 to use our tokenizer.

```
pip install transformers[torch,mistral-common]
```

Examples

- ▶ Helper
- ▶ Text moderation
- ▶ Multimodal moderation (image + text)

Limitations & Ethical Considerations

- **Uneven coverage.** Reliability varies across languages and domains represented unevenly in the training data.
- **Residual label noise.** Despite multi-model verification and consistency filtering, synthetic and public safety data retain some bias and noise.

- **Adversarial / obfuscated inputs** (encoded or

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can reduce reliability.

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